



General pathology

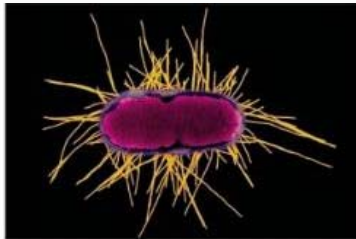
Human Genetics

Lec. 9

Dr.Zainab Hashim

Genetics

- Genetics, the science of heredity, is the study of biological information. All living organisms , from single-celled bacteria and protozoa to multicellular plants and animals, must store and use vast quantities of information to develop, survive, and reproduce in their environments .



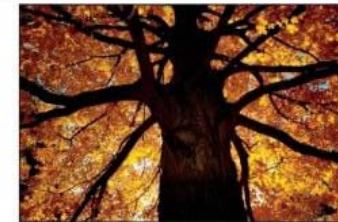
(a) Bacteria



(b) Clown fish



(c) Lion



(d) Oak tree



(e) Poppies



(f) Hummingbird

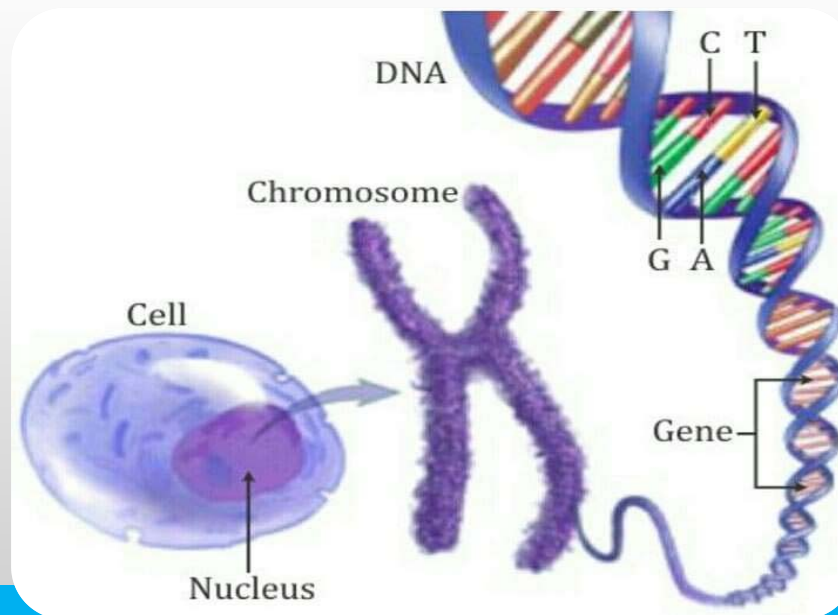


(g) Red-eyed tree frog

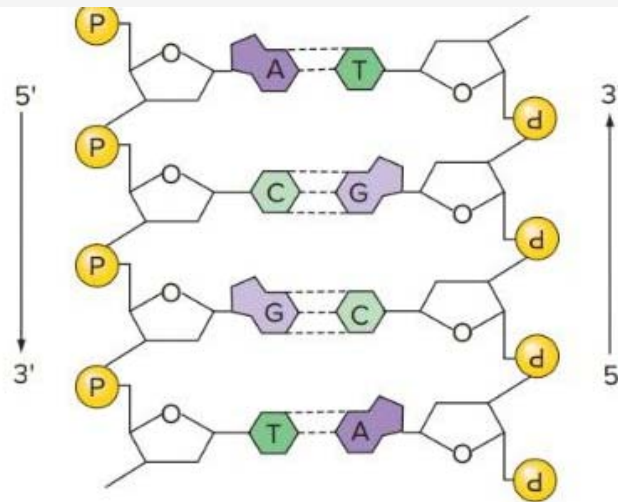


(h) Humans

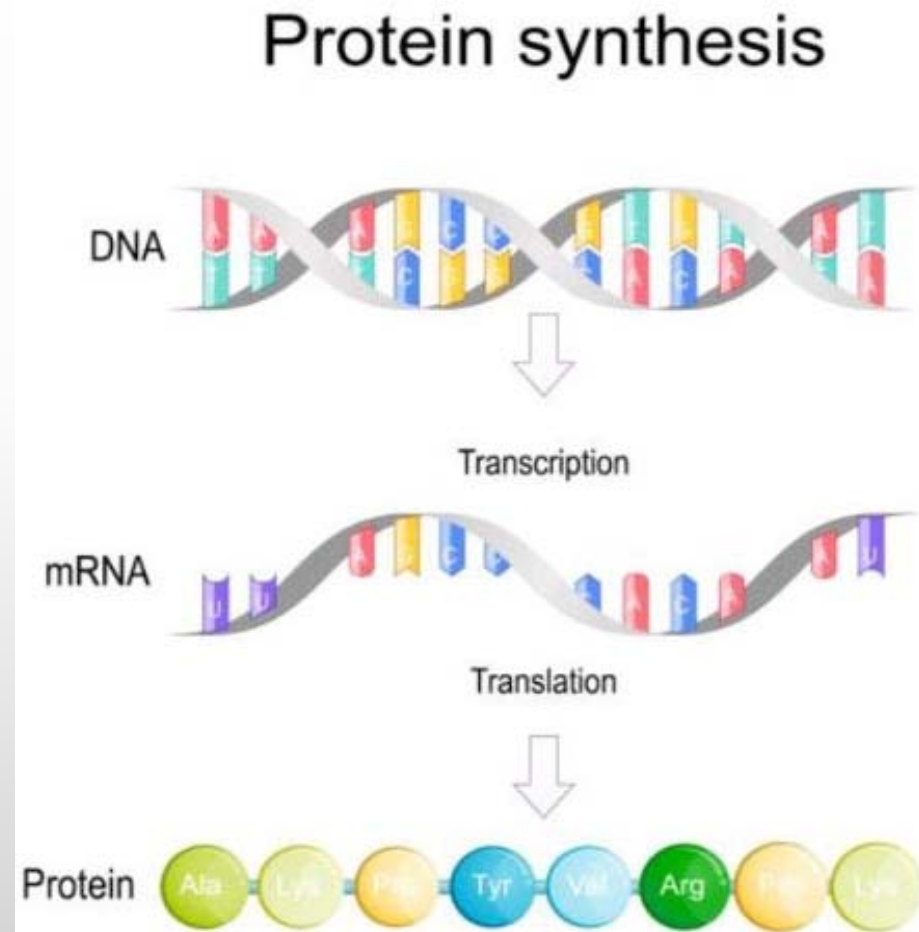
- The linear DNA molecule stores biological information in units known as nucleotides. Within each DNA molecule, the sequence of the four letters of the DNA alphabet, G, C, A, and T, specify which proteins an organism will make as well as when and where protein synthesis will occur.



- The letters refer to the bases, guanine, cytosine, adenine, and thymine , that are components of the nucleotide building blocks of DNA. The DNA molecule itself is a double strand of nucleotides carrying complementary G–C or A–T base pairs .

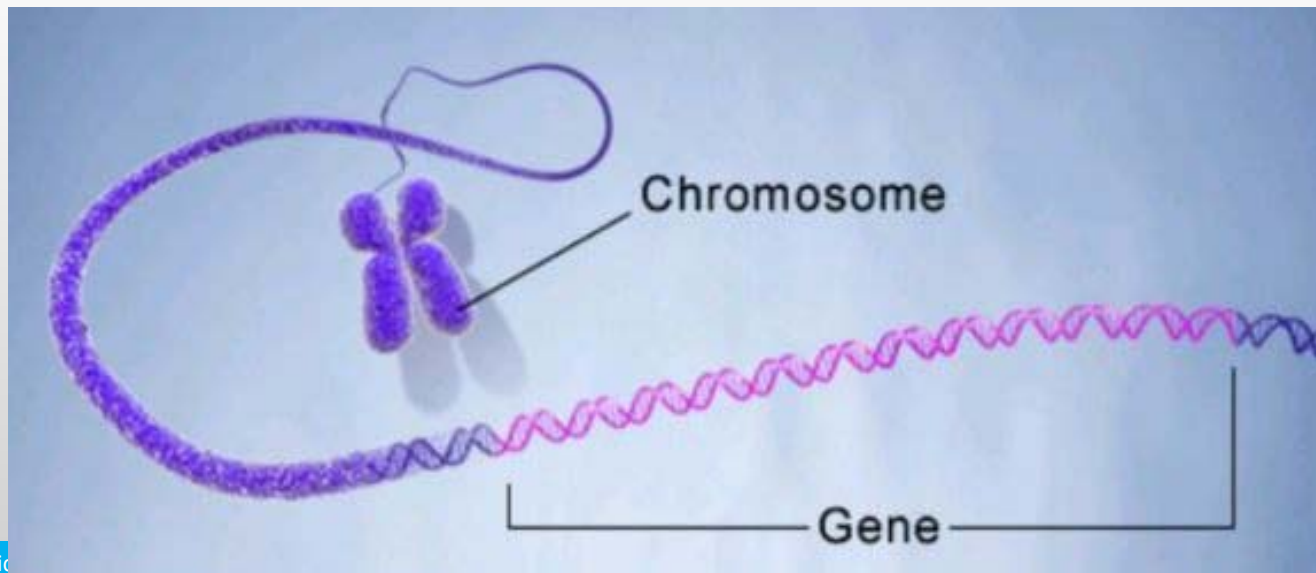


- Most properties of living organisms arise from the class of molecules known as **proteins**, large polymers composed of hundreds to thousands of amino acid linked in a linear sequence.
- The information in the DNA of genes dictates, **via a genetic code**, the order of amino acids in a protein molecule.



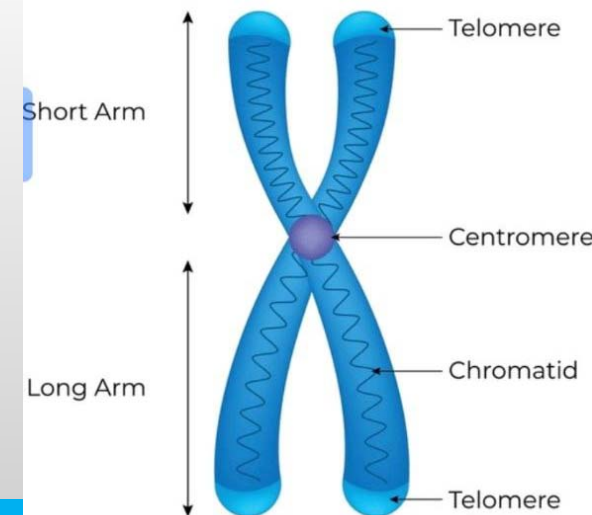
Genes

- **Genes** : the basic unit of inheritance, are contained in chromosomes and consist of deoxyribonucleic acid (DNA). Genes determine the types of proteins and enzymes made by the cell. Thus, genes ultimately influence all aspects of body structure and function. Humans are estimated to have 20,000 to 25,000 genes.

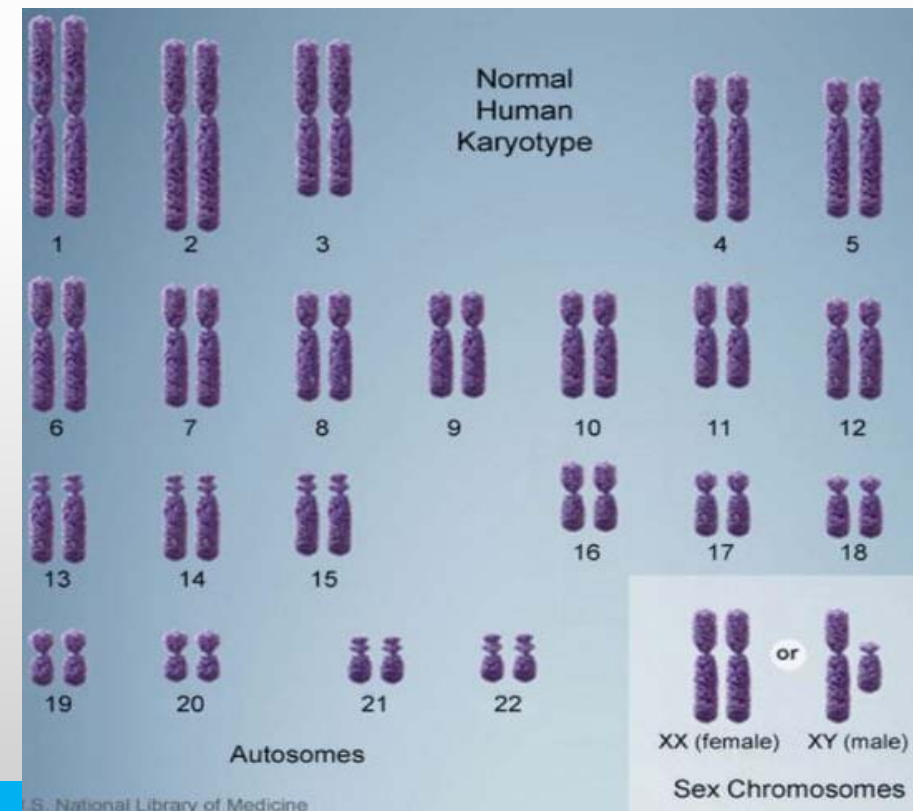


Chromosomes

- Chromosomes: A single piece of coiled DNA and protein that is found in cell nucleus.
- Chromosome forms the classic four arm structure a pair of sister chromatid attached to each other at the centromere.
- **Centromere** is the point where the two chromatids touch and the narrow region that divides the chromosome into short arm and long arm.
- **Chromatid** one of the two identical parts of the chromosome.



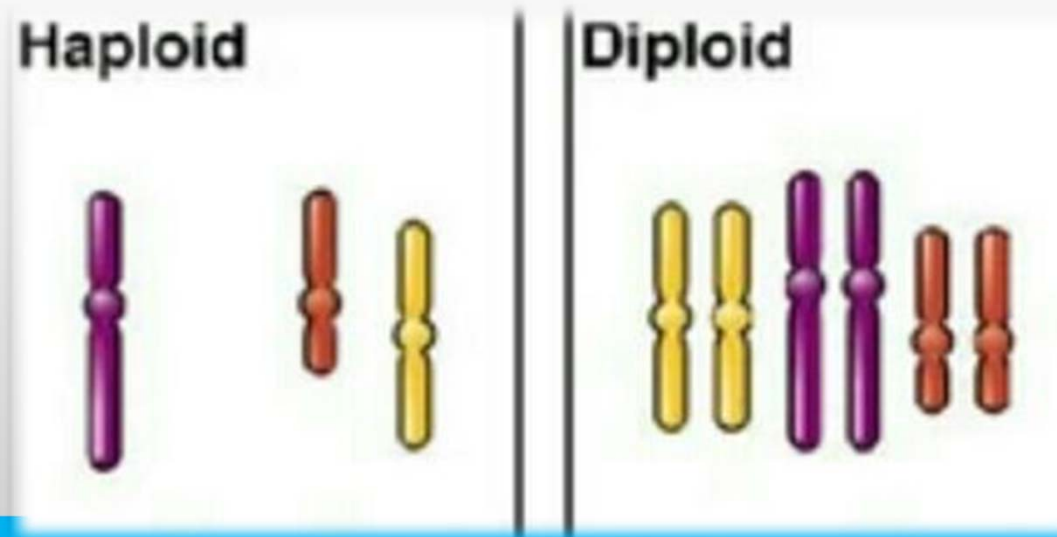
- The chromosomes are arranged in pairs; one member of the pair is inherited from the father, the other from the mother.
- In the human, **46 single** or **23 pairs** of chromosomes are present. Of the 23 pairs of human chromosomes, there are 22 pairs called **autosomes** that are alike in males and females. Each of the 22 pairs of autosomes has the same appearance in all individuals .



- The **sex chromosomes** make up the 23rd pair of chromosomes. Two sex chromosomes determine the sex of a person. All males have an X and Y chromosome (i.e., an X chromosome from the mother and a Y chromosome from the father); all females have two X chromosomes (i.e., one from each parent).

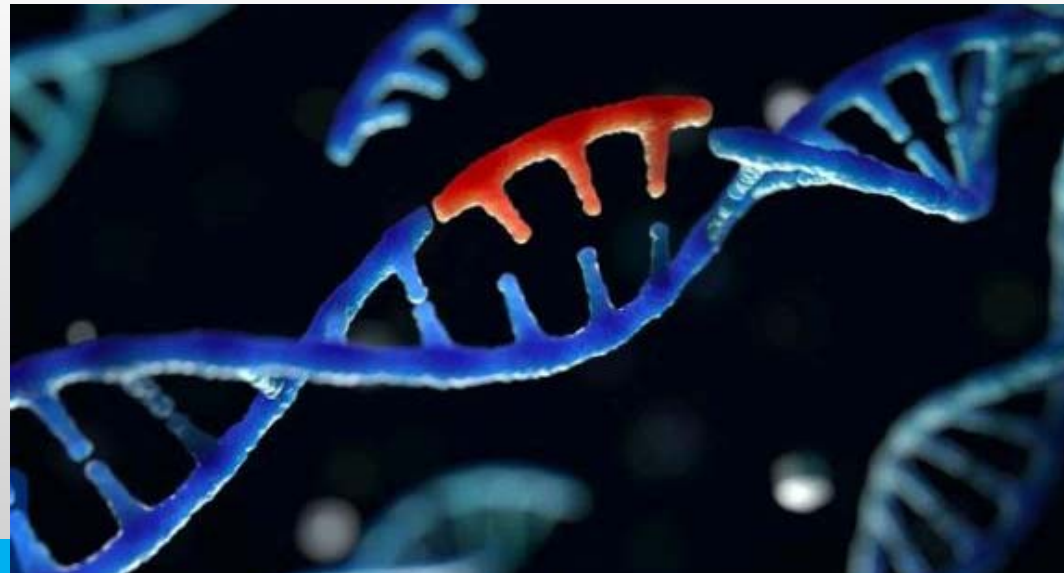


- Human body have somatic cells (body cells), which are **diploid [2n]** having two sets of chromosomes, one from the mother and one from the father. Gametes, reproductive or germ cells, are **haploid [1n]**, they have one set of chromosomes.



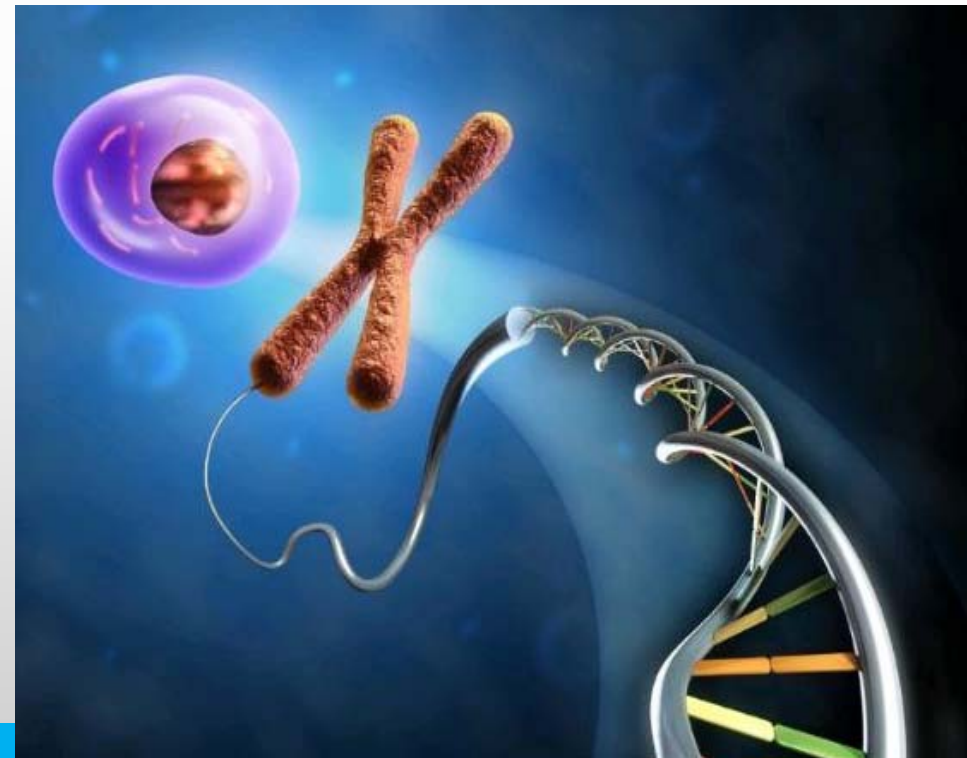
Genetic diseases

A genetic **disease** is a disease that caused by an abnormality in an individual's DNA. Abnormalities can range from a small mutation in a single gene to the addition or subtraction of an entire chromosome or set of chromosomes.



Classification of genetic diseases

- 1) Chromosomal disorders
- 2) Single gene disorders
- 3) Multifactorial disorders

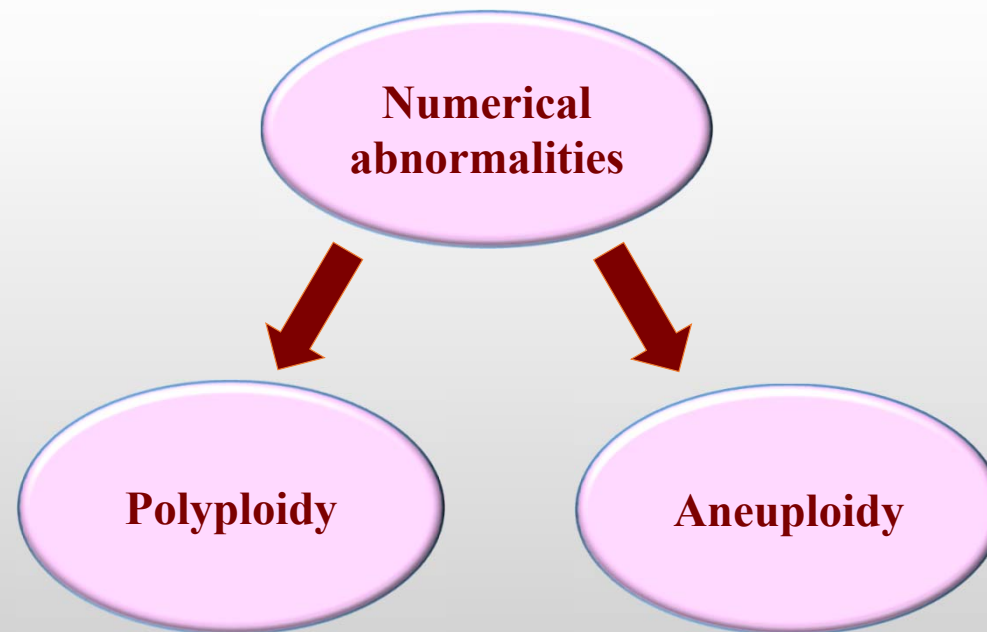


1. Chromosomal disorders

- These are group of genetic diseases arising from chromosomal alteration which are either **numeric** or **structural** abnormalities in the chromosomes that may affect autosomes or sex chromosomes. They form a major category of genetic diseases that are responsible for a large proportion of early abortions, congenital malformations, and mental retardation .

Numerical abnormalities of chromosomes

- It involve the abnormal chromosome number due to an extra or missing chromosome (known as aneuploidy) or the addition of one or more complete haploid complements (known as polyploidy) .

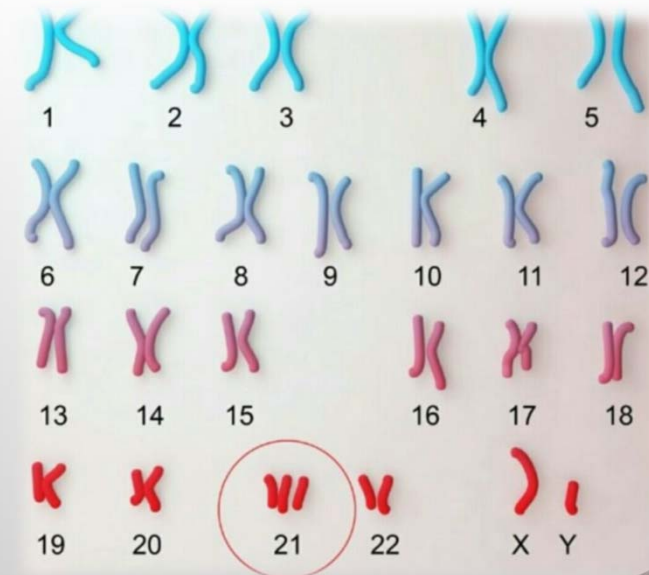


Aneuploidy

- It is abnormal chromosome number due to an extra or missing chromosome that always associated with physical or mental abnormalities or both such as :
- **Trisomy** :having three copies of chromosome instead of normal two (Down syndrome , trisomy 21)
- **Monosomy**

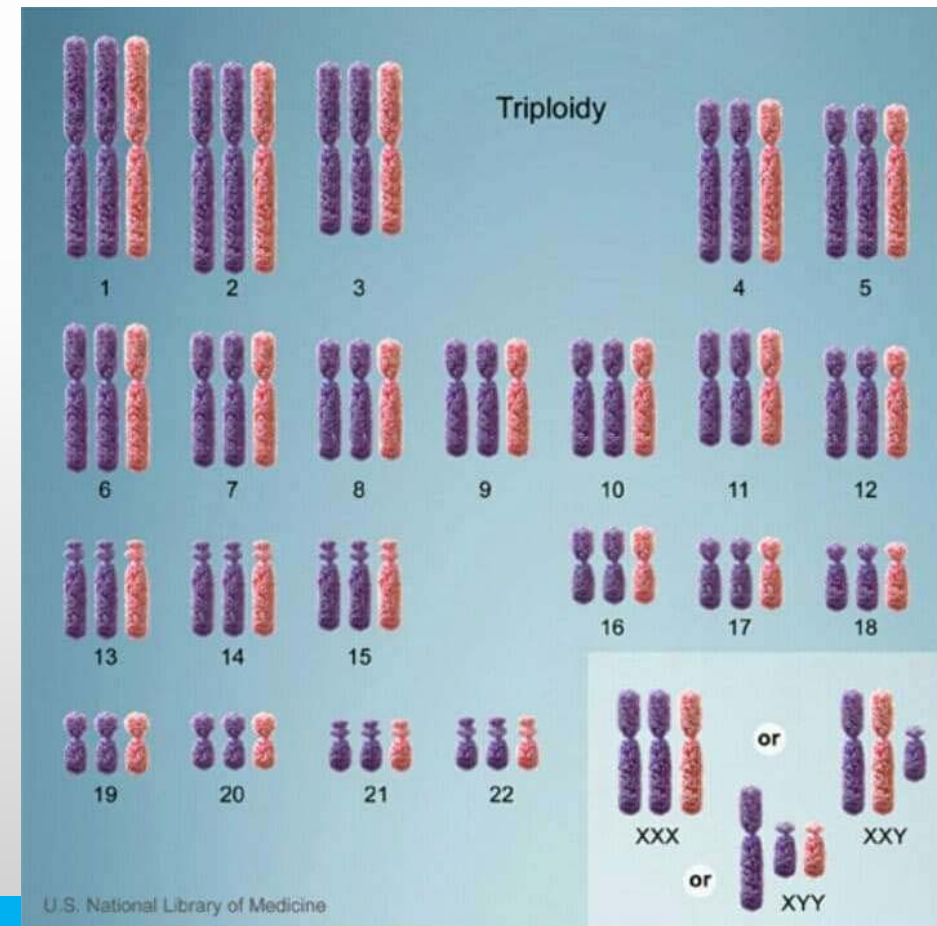


Down syndrome



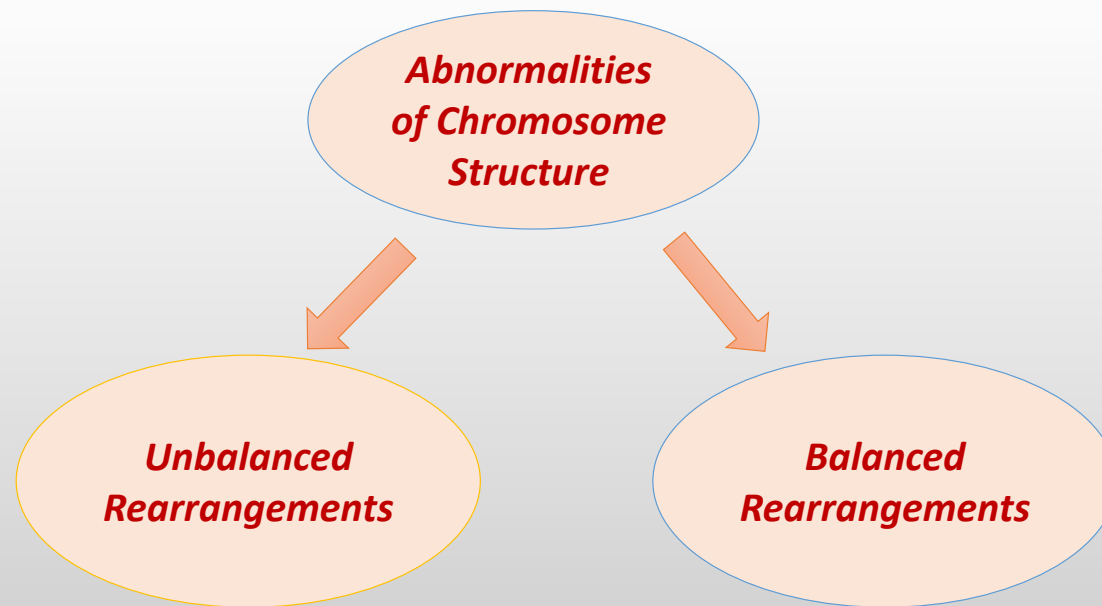
Polyploidy

- Addition of one or more complete haploid complements (multiples of haploid number which are higher than diploid)
- Triploid (3n)
- Tetraploid (4n) and so on .



Structural abnormalities of chromosomes

- Result from chromosome breakage, recombination, or exchange, followed by reconstitution in abnormal combination .

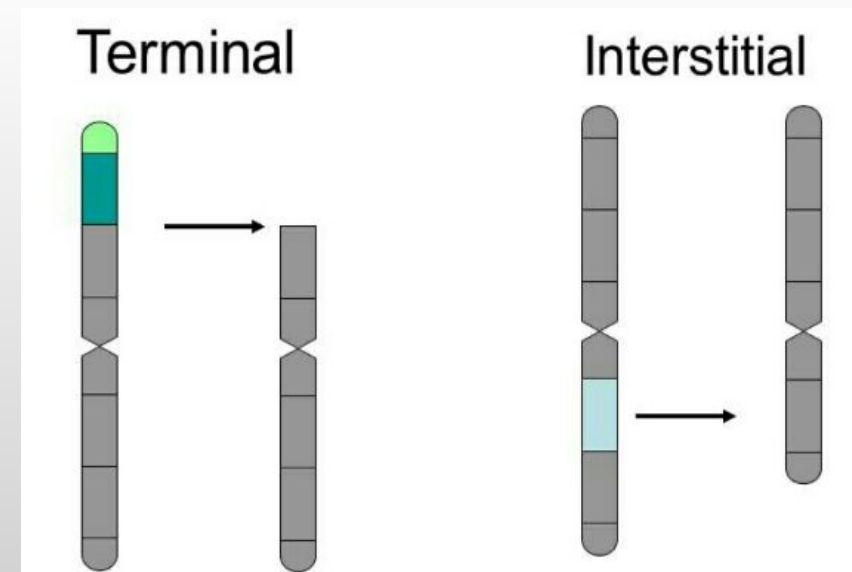


Unbalanced Rearrangements

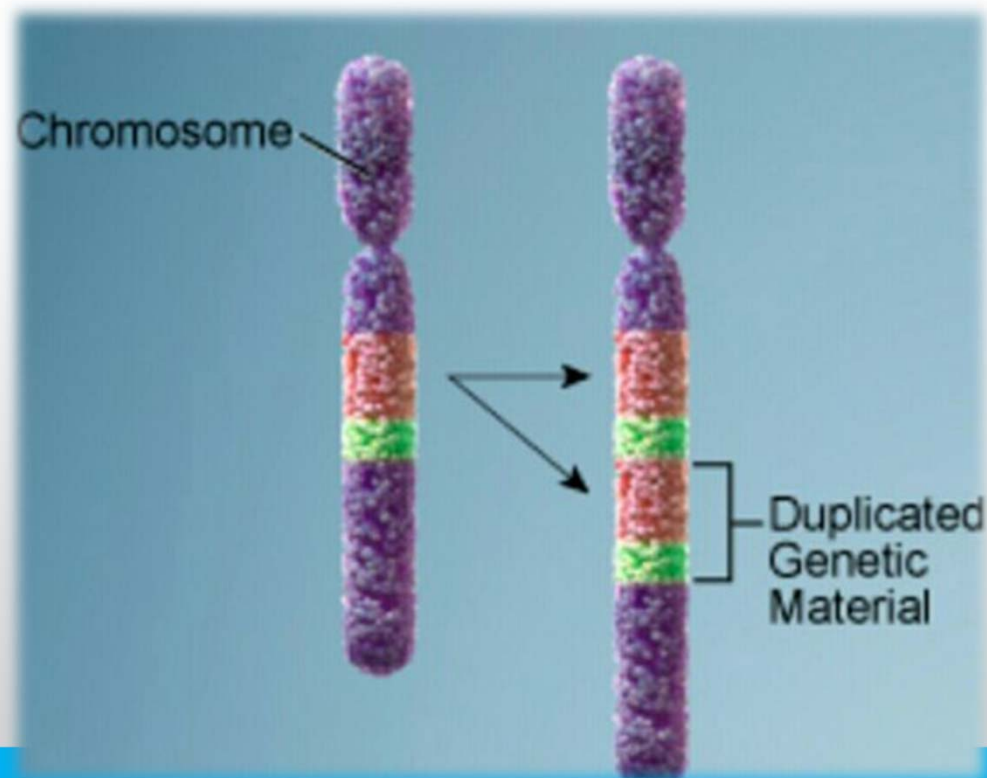
In which the chromosomal complement contain an incorrect amount of chromosomal material (additional or missing material) such as :

Deletions :

1. **Terminal deletion** involve a single break of the terminal part of chromosome and broken part is subsequently lost.
2. **Interstitial deletion** involve two breaks and intervening portion of the chromosome is lost .



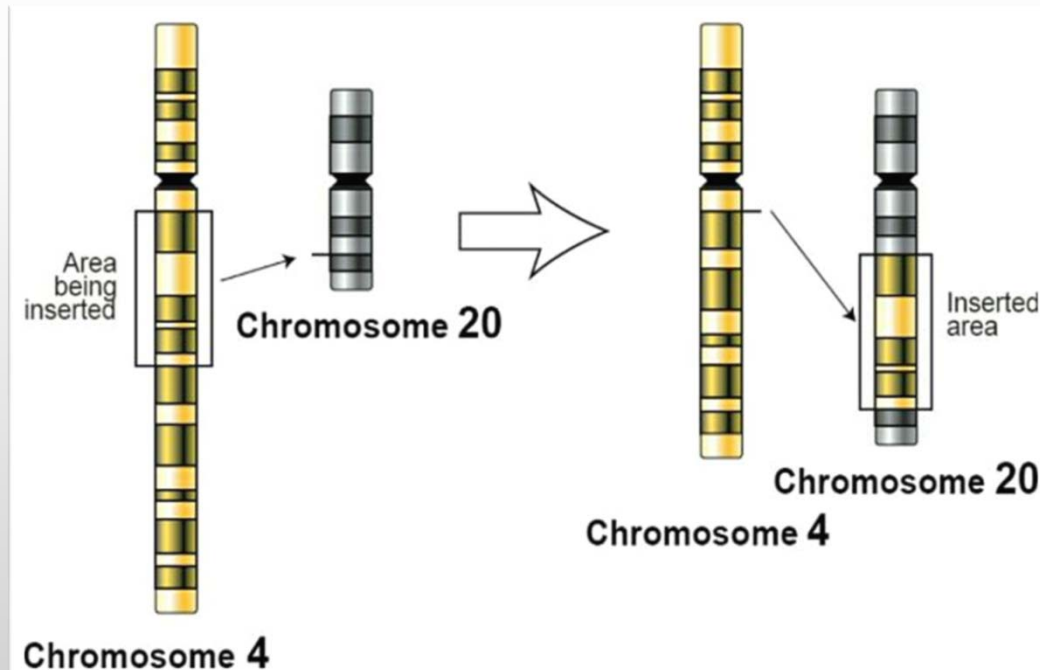
- **Duplication** :some genetic material doubled leads to partial trisomy for the genes within that segment .



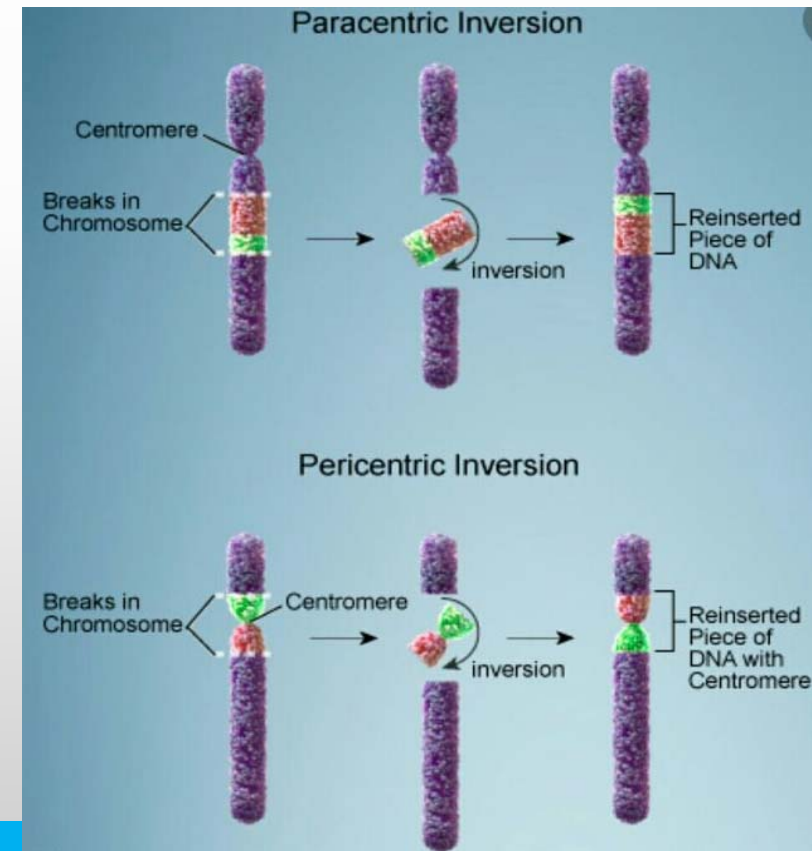
Balanced Rearrangements

The exchange or rearrangement of genetic material does not cause any missing or extra genetic material .

Insertions: occurs when a segment removed from one chromosome is inserted into a different chromosome, either in its usual orientation with respect to the centromere or inverted.

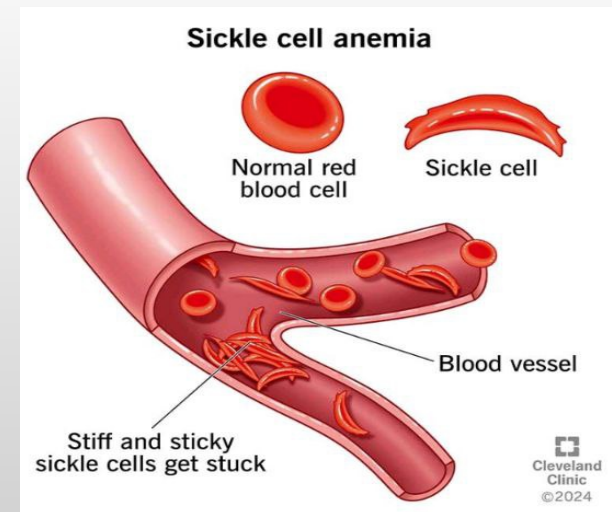
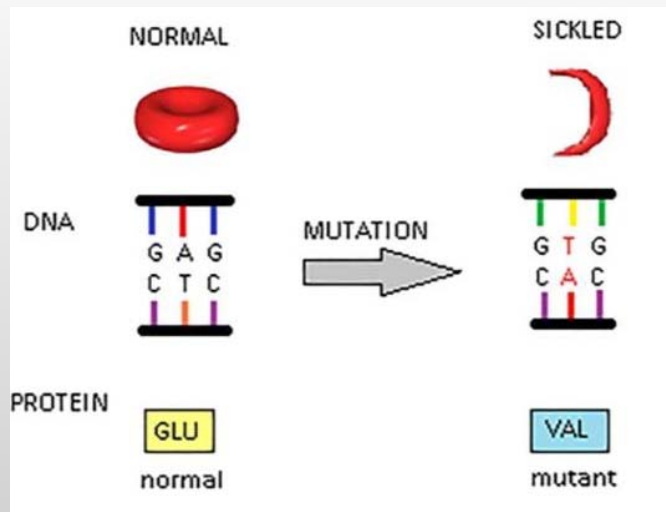


- **Inversions**: an inversion occurs when a single chromosome undergoes two breaks and is reconstituted with the segment between the breaks inverted.
- Inversions are of two types :
 - **Paracentric**, in which both breaks occur in one arm .
 - **Pericentric**, in which there is a break in each arm, around the centromere.



2. Single gene disorders

- Single-gene disorders result when a mutation causes the protein product of a single gene to be altered or missing such as sickle cell disease
- In sickle cell disease: Glutamic acid (normal) → Change to valine (**mutant**) .



3. Multifactorial disorders

- Multifactorial disorders result from mutations in multiple genes, often coupled with environmental causes. such as hypertension, diabetes mellitus .

